

CHAPTER 4. NEURAL NETWORKS

I. Using the TensorFlow Library

Use the Neural Network method to build a classification model. Evaluate the performance of the model on test datasets by the following metrics: Accuracy, Precision, Recall, and F1-score.

Exercise 1

Given the dataset in the file `winedata.csv`, split the dataset into training and test datasets with a training - to - test ratio of 80 : 20.

The feature set includes: `Alcohol`, `Malic acid`, `Ash`, `Alcalinity of ash`, `Magnesium`, `Total phenols`, `Flavanoids`, `Nonflavanoid phenols`, `Proanthocyanins`, `Color intensity`, `Hue`, `OD280/OD315 of diluted wines`, and `Proline`.

The label set is the `Class` column.

Exercise 2

Given the dataset in the file `cars.csv`, split the dataset into training and test datasets with a training - to - test ratio of 70:30.

The feature set includes: `buying`, `maint`, `doors`, `persons`, `lug_boot`, and `safety`.

The label set is the `acceptability` column.

Encode the data in each text column of the training data. Then, use the same encoding scheme to encode the test dataset. For example, if “vhigh” is encoded as 0 and “high” is encoded as 1 in the train dataset, then “vhigh” must also be encoded as 0 and “high” as 1 in the test dataset.

Exercise 3

Given the dataset in the file `bankmarketing.csv`, split the dataset into training and test datasets with a training - to - test ratio of 80 : 20.

The feature set includes: `age`, `marital`, `education`, `housing`, and `duration`.

The label set is the `bankrupt` column.

Encode the data in each text column of the training dataset. Then, use the same encoding scheme to encode the test dataset. For example, if “married” is encoded as 0 and “single” as 1 in the training dataset, then “married” must also be encoded as 0 and “single” as 1 in the test dataset.

II. Developing a Neural Network Class

1. Develop a Neural Network class including: a `fit` function for training the model (calculating the weights (`w`) and bias (`b`)), and a `predict` function for predicting the labels of a new set of samples (i.e., the test dataset).
2. Use the `fit` and `predict` functions to perform the tasks specified in Section I.